

DECODELABS

Industrial Training Kit
Data Analytics Internship

Project 3

SQL Data Analysis

Submitted By:

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1. Objective

The objective of this project is to analyze an e-commerce sales dataset using SQL. The analysis demonstrates fundamental SQL operations including SELECT, WHERE, ORDER BY, GROUP BY, aggregate functions such as COUNT, SUM and AVG, and HAVING. The analysis is used to summarize sales performance and identify useful business insights.

2. Dataset Description

The dataset contains 1,200 e-commerce records and 14 columns. The fields include OrderID, Date, CustomerID, Product, Quantity, UnitPrice, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, ItemsInCart, CouponCode, ReferralSource, and TotalPrice.

The dataset was imported into the SQL Server database named Decodelab_project13 and analyzed using SQL Server Management Studio (SSMS) 22.

3. SQL Operations Performed

SQL Concept	Purpose
SELECT	Retrieve records from the e-commerce table
WHERE	Filter records based on a condition
ORDER BY	Sort records by a selected column
GROUP BY	Summarize records by categories
COUNT()	Count orders/records
SUM()	Calculate total sales
AVG()	Calculate average sales
HAVING	Filter grouped results

4. Basic Filtering and Sorting

A SELECT query retrieved the complete dataset. WHERE filtered Laptop orders and returned 173 records. ORDER BY sorted orders by TotalPrice in descending order so the highest-value orders appeared first.

5. Product-wise Sales Analysis

Product	Orders	Total Sales	Average Sales
Chair	178	195620.1094	1098.9894
Printer	181	195612.6098	1080.7327
Laptop	173	192126.5607	1110.5582
Tablet	179	186568.9505	1042.2846
Monitor	163	175651.4104	1077.6160
Desk	170	167459.9308	985.0584
Phone	156	151722.3899	972.5794

Chair recorded the highest total sales among products at approximately 195,620.11. Phone recorded the lowest total sales at approximately 151,722.39.

6. Payment Method Analysis

Payment Method	Orders	Total Sales	Average Sales
Online	258	262442.9406	1017.2210
Cash	246	259786.2894	1056.0418
Credit Card	234	263847.6308	1127.55398
Debit Card	232	232361.1807	1001.5568
Gift Card	230	246323.9202	1070.9736

Online had the highest order count with 258 orders. Credit Card generated the highest total sales among payment methods, at approximately 263,847.63.

7. Order Status Analysis

Order Status	Orders	Total Sales	Average Sales
Cancelled	250	276396.2095	1105.5848
Returned	247	243277.7005	984.9299
Pending	237	256328.1503	1081.5534
Shipped	235	246159.5807	1047.4878
Delivered	231	242600.3207	1050.2174

Cancelled was the largest order-status group with 250 orders and also had the highest total sales in this analysis, at approximately 276,396.21.

8. Referral Source Analysis

Referral Source	Orders	Total Sales	Average Sales
Instagram	259	275285.4506	1062.8782
Email	250	261808.5511	1047.2340
Google	241	250441.4802	1039.1766
Facebook	228	250410.8990	1098.2934
Referral	222	226815.5792	1021.6918

Instagram was the strongest referral source by both order count and total sales, with 259 orders and approximately 275,285.45 in total sales.

9. HAVING Analysis

The HAVING query identified products whose grouped total sales were greater than 180,000.

Product	Orders	Total Sales
Chair	178	195620.1094
Printer	181	195612.6098
Laptop	173	192126.5607
Tablet	179	186568.9505

10. Additional Analysis

A TOP 5 query was executed to identify the five highest-value individual orders using OrderID, Product, Quantity, UnitPrice, PaymentMethod, OrderStatus and TotalPrice, sorted by TotalPrice descending.

11. Key Business Insights

- The dataset contains 1,200 e-commerce orders.
- Laptop filtering returned 173 orders.
- Chair had the highest total sales among products, at approximately 195,620.11.
- Credit Card transactions produced the highest total sales among payment methods, at approximately 263,847.63.
- Cancelled orders were the largest order-status group, with 250 orders.
- Instagram was the strongest referral source by both order count and total sales.
- Four products exceeded 180,000 in total sales: Chair, Printer, Laptop and Tablet.

12. Conclusion

This project provided practical experience in SQL-based data analysis using SQL Server Management Studio. The analysis demonstrated how SQL can transform raw transactional data into structured summaries and business insights. The project strengthened skills in data retrieval, filtering, sorting, grouping, aggregation and interpretation of results.

13. Tools Used

Category	Details
Database	Microsoft SQL Server
IDE	SQL Server Management Studio 22
Dataset	E-commerce sales dataset (1,200 records)
Analysis	SELECT, WHERE, ORDER BY, GROUP BY, COUNT, SUM, AVG and HAVING